Advanced Applied Statistics
Lecture 1

Christian Hardmeier

29 August 2025

IT UNIVERSITY OF COPENHAGEN

Practical Matters

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Your Teachers



Christian Hardmeier



Sami Brandt

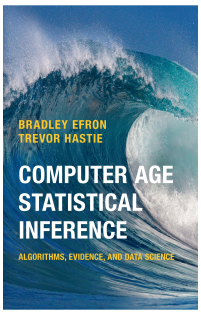
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Learning Objectives

After the course, the student should be able to

- analyse statistical problems and reason about the most appropriate methods to apply;
- apply and reflect on advanced applied statistical methods;
- identify and describe problems that can be solved using multivariate techniques;
- implement basic statistical algorithms and interpret results; and
- summarise the results of an analysis in a statistical report.

Course literature



Bradley Efron and Trevor Hastie:
Computer Age Statistical Inference.

`https://hastie.su.domains/CASI/`

- Also available in hardcopy from Cambridge University Press.
- Note: Due to added exercises, the page numbers in digital edition and 2021 hardcopy edition don't match, even though the text is identical.
- We'll be careful to refer to sections, not pages.
- *Notes and Details* sections are optional.
- Additional reading assignments on LearnIT.

Course overview

			CASI
26/08	Introduction, Frequentist inference	CH	1 and 2
02/09	Maximum likelihood, generalised linear models	CH	4 and 8
09/09	Parametric models, Exponential families	SB	5
16/09	Bayesian inference, Empirical Bayes	SB	3 and 6
23/09	Sparse modelling, Lasso, Ridge regression	CH	7 and 16
30/09	Survival analysis, Expectation-Maximisation	SB	9
07/10	Jackknife and bootstrap	SB	10
14/10	*Midterm break, no meetings*		
21/10	Bootstrap confidence intervals	SB	11
28/10	Cross-validation	SB	12
04/11	Large-scale hypothesis testing	CH	15
11/11	Objective Bayes inference and MCMC	SB	13 and 14
18/11	*to be determined*	SB	*tbd*
25/11	*Project supervision*		
02/12	*Project supervision*		

Getting help

- **Technical questions** or **course housekeeping**:
  Please use the **forum on LearnIT**!
- **Formal accommodations** (e.g., deadline extensions):
  Get in touch with SAP (sap@itu.dk).
- Sensitive personal feedback or requests:
  Email us or use the anonymous web form.
- And: Come to lectures, tutorials, study lab
  and talk to us!

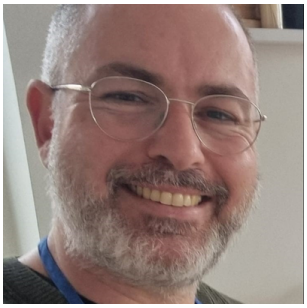
Giving us feedback

- Is there something we can improve?
- Did we say something that made you feel uncomfortable?
- Did something happen you think we need to address in class?

You can always

- write to Christian (chrha@itu.dk) or Sami (sambr@itu.dk)
- or use the anonymous feedback form:
  `https://forms.cloud.microsoft/e/trLLRv6Gtu`

Your TAs



Sa'ar



Tonio

Exercises

- Tutorial/exercises each week immediately after the lecture, in room 5A60.
- The weekly exercises may be split in multiple parts.
- Starting in the second week, some exercises will be highlighted as **particularly relevant for the exam**.
- Lab sessions will consist of
  - getting started with current week's exercises,
  - a short recap of the previous week's lecture, and
  - discussion of the previous week's exercises.
- No submission or grading of exercises, but the TAs are available for **individual help and feedback** in **room 3A18 on Tuesdays after 16:00**.

Today's exercise

1. **Probability gymnastics**
   A set of exercises on elementary probability theory.
   Please do these exercises and read up on anything that you don't master.
2. **Regression exercise**
   A small exercise involving regression on a dataset from the course book.
   Just to get you into the mood.

Examination

- The course is examined in the form of a *project* and an *oral exam*.
- Assignment: Statistical analysis on a dataset of your choice using methods we discussed.
- Group formation before midterm break (but start thinking about it as you follow the course).
- More specific instructions will follow.
- Submission deadline: **Fri 2 Jan 2026, at 2 pm**.
  (Any extension must be approved by SAP.)

Oral Exam

- Oral exam in project groups on **20–22 January 2026**.
  1. Joint presentation of project.
  2. Questions to whole group.
  3. Questions to students individually.
- Scope of the exam:
  - Project and project report.
  - Course in general
    (Syllabus will be available before the last lecture).
- Your best preparation is to **do the exercises religiously!**
- Put yourself in a good starting position with a nice project!

Statistical inference

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Business questions

Statistical inquiry is usually motivated by a business problem.

- How much profit can our toy shop expect to make next December?
- How many doctors do we need to hire for adequate staffing of an emergency room on Friday night?
- Does our new machine translation system output better translations than the old one?
- Does this batch of apple cider vinegar meet the necessary quality standards?
- How many respondents do we need to interview to predict the outcome of the next election sufficiently well?

Prediction **Decision problems** **Experimental design**

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Statistical model (informally)

- Mathematical model covering the statistical assumptions around the generation of sample data.
- Idealised representation of the data generation process.
- Reduced, simplified description of reality that serves as the basis for statistical inference.

Experiment and Event

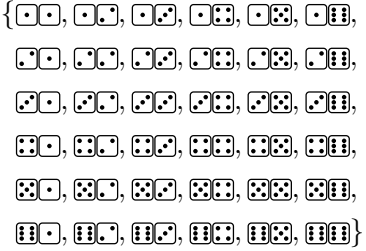
An experiment is any process, real or hypothetical, in which the possible outcomes can be identified ahead of time.

An event is a well-defined set of possible outcomes of the experiment.

Sample space

The sample space is the set of all possible outcomes.

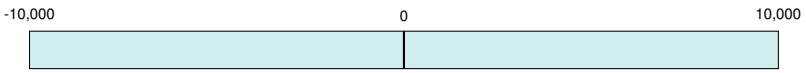
Example:
Casting a pair of dice:

$$S = \{$$

$$\}$$

Sample space

The sample space is the set of all possible outcomes.

Example:
Profit of a shop



Events

An event is a subset of the sample space.

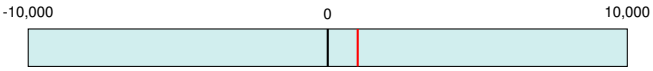
- The first die is a 3, and the second is a 6.

$$\{\boxdot\boxdot\}$$

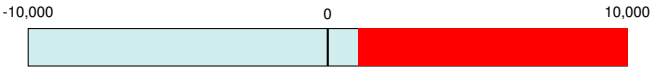
- The dice sum to 6.

$$\{\boxdot\boxdot, \boxdot\boxdot, \boxdot\boxdot, \boxdot\boxdot, \boxdot\boxdot\}$$

- The profit is 1,000 DKK.



- The profit is greater than 1,000 DKK.



Random variable

A real-valued function $X : S \mapsto \mathbb{R}$ defined on the sample space.

Example: Sum of the dice

$$X(\boxdot\boxdot) = 2$$
$$X(\boxdot\boxdot) = 3$$
$$X(\boxdot\boxdot) = 4$$
$$\dots$$
$$X(\boxdot\boxdot) = 7$$
$$X(\boxdot\boxdot) = 8$$
$$X(\boxdot\boxdot) = 9$$
$$\dots$$

What do we want to know about random variables?

Full specification

- What values can they assume, and how likely are they?
  Distribution

Descriptive statistics

- What is their value typically?
  Central tendency: Expectation/mean, mode, median, ...
- How certain are we about this typical value?
  How "spread out" are the values?
  Dispersion: Variance, standard deviation, interquartile range, ...

Statistical model

- *Scope: What do we want to model?*
  Identification of **random variables of interest**
  (observable/hypothetically observable)
- *Structure: How does it hang together?*
  Specification of **joint distribution of random variables**
- *Parameters: How can we fit it?*
  Identification of **parameters of distribution**
  assumed unknown
- *Optionally: What parameter values are likely?*
  Specification of **joint distribution for unknown parameters**
  if parameters are treated as random variables

Frequentist inference

Frequentist inference

The frequentist approach to statistics assumes that

- experiments are infinitely repeatable, and
- the underlying parameters remain constant.

Probabilities can then be estimated as *relative frequencies* in a long run of experiments.

A typical quantity of interest is the standard error of an estimate:
This is the *standard deviation of the sampling distribution* of a statistic.

Sample statistic vs. estimator

- Let $\mathbf{X} \sim F$ be a random variable with an unknown distribution, and $\mathbf{x}$ be a sampled realisation.
- We wish to estimate a parameter θ of $F = F(\theta)$.

$$\hat{\theta} = t(\mathbf{x}) \qquad\qquad \hat{\Theta} = t(\mathbf{X})$$

- $\hat{\theta}$ is a *sample statistic*: a number computed from the sample.
- $\hat{\Theta}$ is a *random variable*: an estimator of $\hat{\theta}$.
- Note: In this paradigm, the parameter itself is not a random variable, but its estimator is.
- The sampling distribution of the estimator quantifies our uncertainty about the parameter.

Accuracy of an estimate

The accuracy of an observed estimate $\hat{\theta} = t(\mathbf{x})$ is the probabilistic accuracy of $\hat{\Theta} = t(\mathbf{X})$ as an estimator of θ.

- Let μ be the expectation the expectation of $\hat{\Theta}$:

$$\mu = E_F(\hat{\Theta})$$

- **Bias:** How far off is the estimator on average?

$$\text{Bias} = \mu - \theta$$

- **Variance:** What is the spread of our estimates?

$$\text{Variance} = E_F\left((\hat{\Theta} - \mu)^2\right)$$

Frequentism in practice

- *The probabilistic properties of a procedure of interest are derived and then applied verbatim to the procedure's output for the observed data.*
- In deriving the inference procedures, we assume we know the data distribution F, but in reality we don't.
- Various tools help us get around this problem.

Plug-in principle

The frequentist accuracy estimate of the data is itself estimated from the observed data.

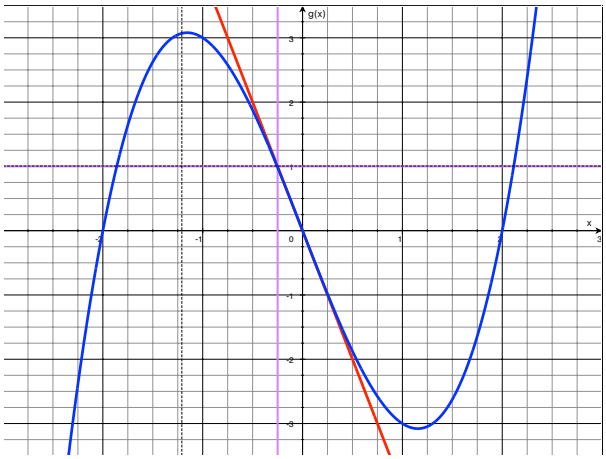
Example:

- Let p be the observed proportion of heads in n flips of a (biased) coin having true probability π.
- The true standard error of p is
$$\sqrt{\frac{\pi(1-\pi)}{n}}.$$
- We don't know π, so we can only report
$$\sqrt{\frac{p(1-p)}{n}}.$$

Delta method and Taylor-series approximations

- If there is a sequence of random variable satisfying
$$\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \sigma^2),$$
where θ and σ^2 are finite constants and $\xrightarrow{d}$ denotes *convergence in distribution*, then
$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{d} N(0, \sigma^2 \cdot [g'(\theta)]^2)$$
under mild assumptions on g.
- The first assumption is true whenever the central limit theorem holds.
- This allows us to estimate parameters of transformed quantities.

Delta method and Taylor-series approximations



$$g(x) \approx g(\theta) + g'(\theta)(x - \theta)$$

If the slope is large, small variation in x translates into large variation in $g(x)$.

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Precision and recall

Predicted labels

Gold labels	ADJ	ADP	ADV	CONJ	DET	NOUN	PRON	PROPN	PUNCT	VERB	X
ADJ	1385	4	63	0	0	243	0	0	0	88	1
ADP	4	1734	15	2	0	15	0	0	0	3	248
ADV	62	97	906	0	29	71	0	0	0	10	27
CONJ	1	387	71	761	12	3	0	0	0	3	8
DET	9	3	3	1	2228	17	11	0	0	1	0
NOUN	109	3	6	0	20	3965	1	0	2	85	5
PRON	23	30	6	0	204	155	1779	0	0	21	1
PROPN	35	0	2	1	4	1811	2	0	0	19	5
PUNCT	37	9	8	1	5	263	0	0	2783	45	7
VERB	51	10	8	0	2	243	0	0	0	3954	4
X	36	15	192	1	18	386	20	0	4	120	436

Precision and recall

Predicted labels

Gold labels	NOUN	PRON	PROPN	PUNCT	VERB	X
NOUN	3965	1	0	2	85	5
PRON	155	1779	0	0	21	1
PROPN	1811	2	0	0	19	5
PUNCT	263	0	0	2783	45	7
VERB	243	0	0	0	3954	4
X	386	20	0	4	120	436

$$\text{Precision} = \frac{\text{diagonal element}}{\text{column sum}}$$

$$= \frac{3965}{7172} = 0.553$$

Precision and recall

Predicted labels

Gold labels						NOUN	PRON	PROPN	PUNCT	VERB	X
						243	0	0	0	88	1
						15	0	0	0	3	248
						71	0	0	0	10	27
						3	0	0	0	3	8
						17	11	0	0	1	0
NOUN	109	3	6	0	20	3965	1	0	2	85	5
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VERB	51	10	8	0	2	243	0	0	0	3954	4
X	36	15	192	1	18	386	20	0	4	120	436

$$\text{Recall} = \frac{\text{diagonal element}}{\text{row sum}}$$

$$= \frac{3965}{4196} = 0.945$$

Sampling distribution of precision and recall

- Why do we need the sampling distribution?
  $\rightarrow$ tests, confidence intervals, etc.
- (Observed) precision is the mean of a set of indicator variables:

$$P = \frac{1}{N_{\text{pred } c}} \sum_{i=1}^{N_{\text{pred } c}} P_i, \qquad \text{where } P_i = \begin{cases} 1 & \text{if item } i \text{ is of class } c \\ 0 & \text{otherwise} \end{cases}$$

  and the sum is over the examples *predicted* as class c.
- If the $P_i \sim \text{Bernoulli}(p)$, then the sampling distribution will be asymptotically normal by the Central Limit Theorem:

$$P \sim N\left(p, \frac{p(1-p)}{N_{\text{pred } c}}\right)$$

- Analogous argument for recall.

F-score

- We often want to summarise Precision and Recall in a single number.
- **F-score** (or F_1) is the *harmonic mean* of Precision and Recall.

$$F = 2 \cdot \frac{P \cdot R}{P + R}$$

- If P and R are equal, then F is the same.
- If they are different, then F is closer to the *lower* of them.
- By maximising F-score, we emphasise balanced P and R.

Multivariate Delta method

If we have a sequence of p-dimensional vectors $\boldsymbol{T} \in \mathbb{R}^p$ converging in distribution to $\boldsymbol{\theta} \in \mathbb{R}^p$ with asymptotic covariance matrix $\boldsymbol{\Sigma}/n$,

$$\sqrt{n}(\boldsymbol{T}_n - \boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}).$$

The Delta method approximates the variance of a transformed quantity $f(\boldsymbol{T}_i)$, $f : \mathbb{R}^p \mapsto \mathbb{R}$.

$$\mathrm{Var}(f(\boldsymbol{T})) \approx \frac{1}{n} \sum_{j=1}^{p} \sum_{k=1}^{p} \frac{\partial f}{\partial \theta_j} \frac{\partial f}{\partial \theta_k} \sigma_{jk}$$

If $\boldsymbol{T}$ is uncorrelated ($\boldsymbol{\Sigma}$ is diagonal), this simplifies to

$$\mathrm{Var}(f(\boldsymbol{T})) \approx \frac{1}{n} \sum_{j=1}^{p} \left(\frac{\partial f}{\partial \theta_j} \right)^2 \sigma_{jj}$$

Sampling distribution of F-score

- The F-score $F(P, R) = \dfrac{2PR}{P + R}$ is a transformation of the 2-dimensional statistic (P, R).

$$\mathrm{Var}(F(P, R)) \approx \frac{1}{N} \left(\left(\frac{\partial F}{\partial P} \right)^2 \mathrm{Var}(P) + \left(\frac{\partial F}{\partial R} \right)^2 \mathrm{Var}(R) \right)$$

- Assuming again $P_i \sim \mathrm{Bernoulli}(p)$ and $R_i \sim \mathrm{Bernoulli}(r)$, then

$$\mathrm{Var}(P) = p(1 - p) \qquad \text{and} \qquad \mathrm{Var}(R) = r(1 - r)$$

We don't know p and r, but we can use the observed P and R as plug-in estimates.

Sampling distribution of F-score

- The partial derivatives are

$$\frac{\partial F}{\partial P} = \frac{2R^2}{P + R} \qquad \text{and} \qquad \frac{\partial F}{\partial R} = \frac{2P^2}{P + R}$$

- Putting things together, we find

$$\begin{aligned} \mathrm{Var}(F(P, R)) &\approx \frac{1}{N} \left(\left(\frac{\partial F}{\partial P} \right)^2 \mathrm{Var}(P) + \left(\frac{\partial F}{\partial R} \right)^2 \mathrm{Var}(R) \right) \\ &= \frac{1}{N} \left(\left(\frac{2R^2}{P + R} \right)^2 P(1 - P) + \left(\frac{2P^2}{P + R} \right)^2 R(1 - R) \right) \end{aligned}$$

Simulation and the bootstrap

- Modern computer power allows us to simulate the “infinite” sequence of experiments numerically.
- Create B new *bootstrap* samples from your existing data sample by resampling with replacement.
- Run the inference procedure for each of the bootstrap samples.
- Draw conclusions from the empirical distribution of the estimates.